

Chapter 49

Nervous Systems

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

Neil Campbell and Jane Reece

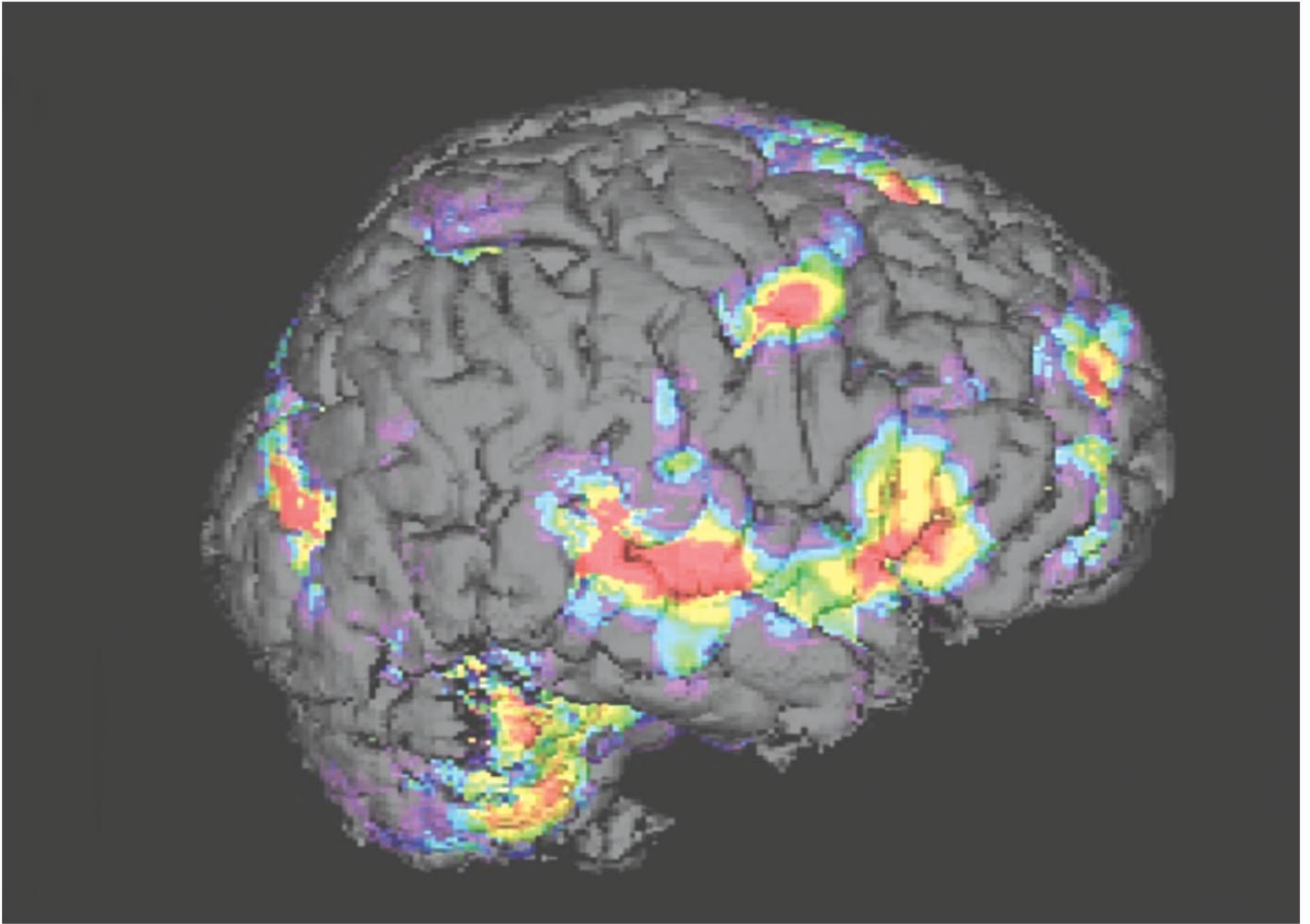
Lectures by Chris Romero, updated by Erin Barley with contributions from Joan Sharp

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Overview: Command and Control Center

- The circuits in the brain are more complex than the most powerful computers.
- Functional magnetic resonance imaging (MRI) can be used to construct a 3-D map of brain activity.
- *The vertebrate brain is organized into regions with different functions.*

Scientists map activity within the human brain

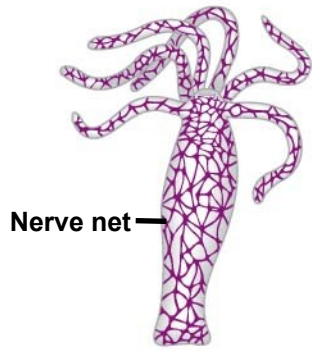


Nervous systems consist of circuits of neurons and supporting cells

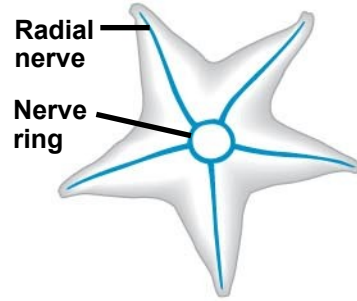
- The simplest animals with nervous systems, the **cnidarians**, have neurons arranged in nerve nets.
- A **nerve net** is a series of interconnected nerve cells. There is no central pathway / or directional organization.
- More complex animals have nerves.

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- **Nerves** are bundles that consist of the **axons** of multiple **nerve cells**.
 - Sea stars have a nerve net in each arm connected by radial nerves to a central nerve ring.

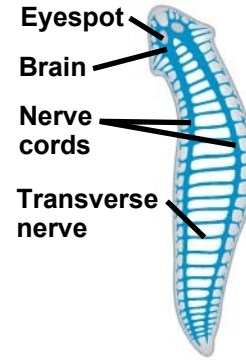
Nervous system organization



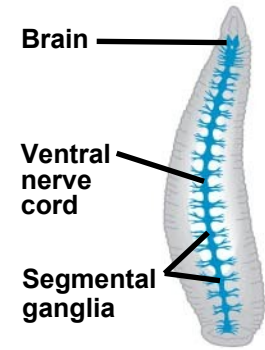
(a) **Hydra** (cnidarian)



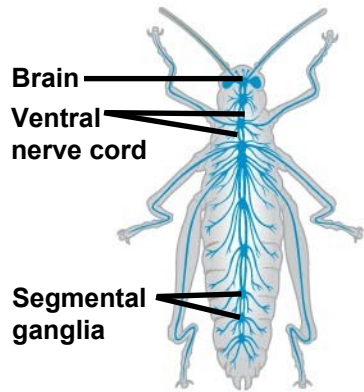
(b) **Sea star** (echinoderm)



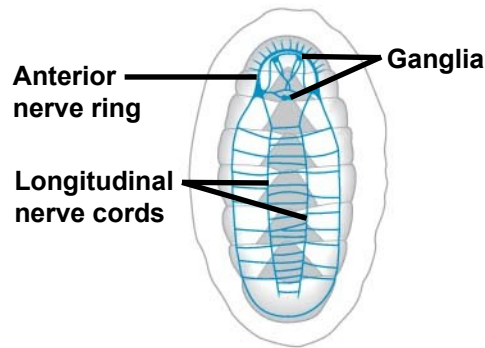
(c) **Planarian** (flatworm)



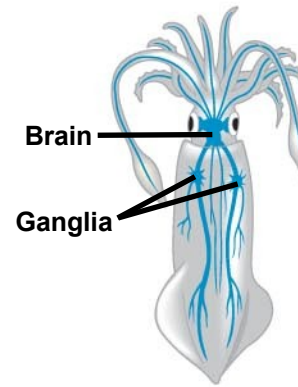
(d) **Leech** (annelid)



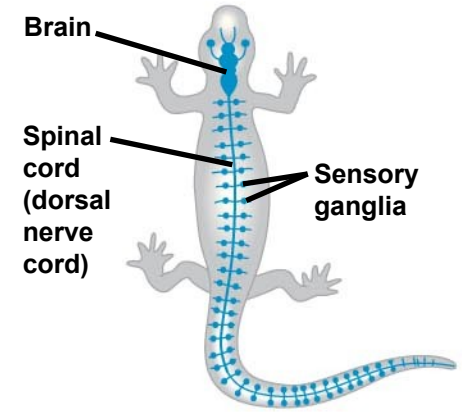
(e) **Insect** (arthropod)



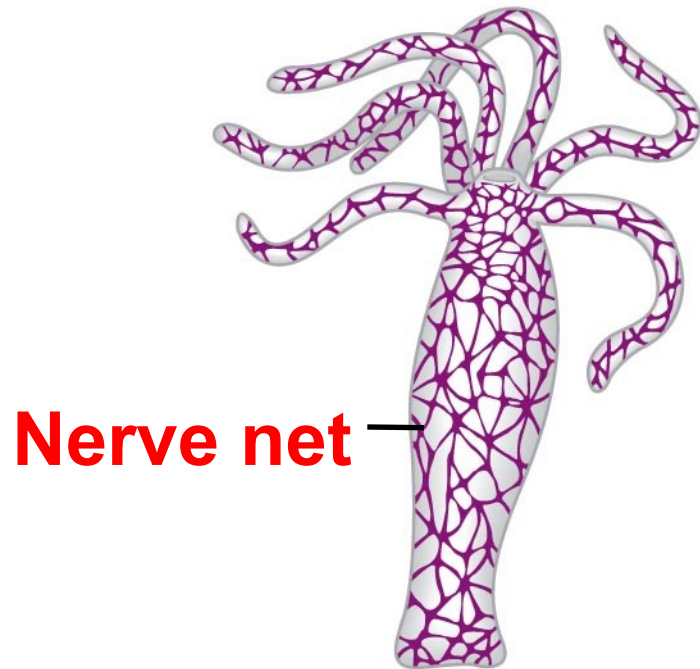
(f) **Chiton** (mollusc)



(g) **Squid** (mollusc)

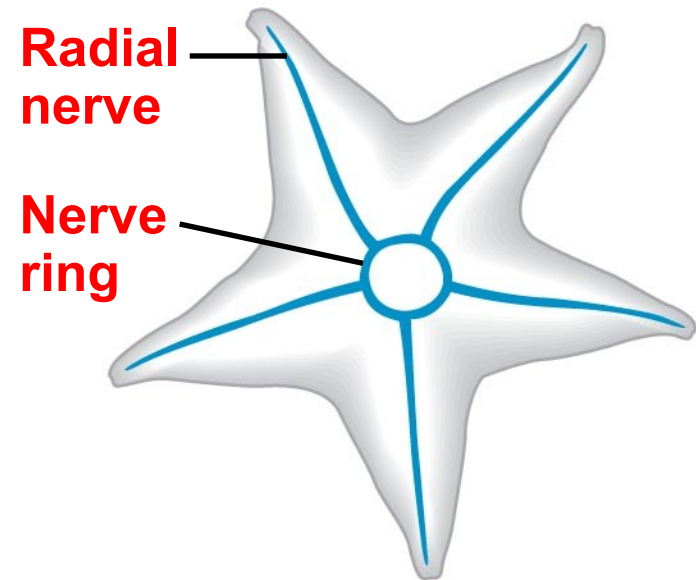


(h) **Salamander** (vertebrate)



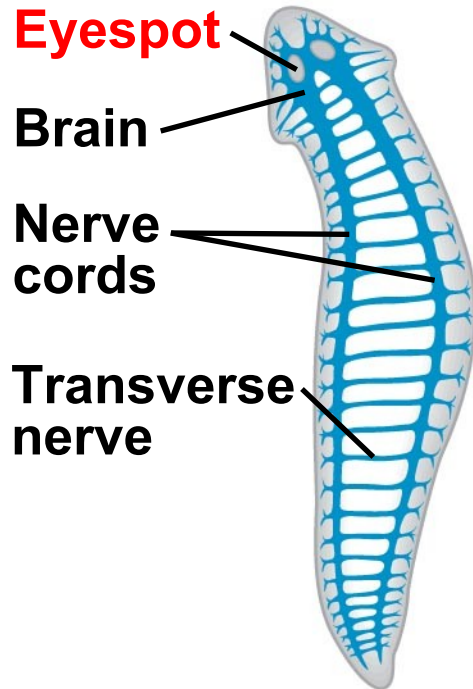
Hydra (cnidarian)

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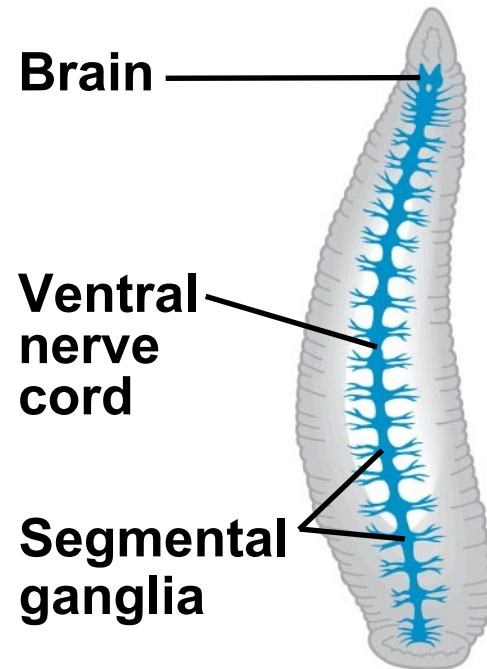
Sea star (echinoderm)

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- **Bilaterally symmetrical** animals exhibit cephalization.
 - **Cephalization** is the clustering of sensory organs at the front end of the body.
 - Relatively simple cephalized animals, such as flatworms, have a *central nervous system (CNS)*.
 - The **CNS** consists of a brain and longitudinal nerve cords.

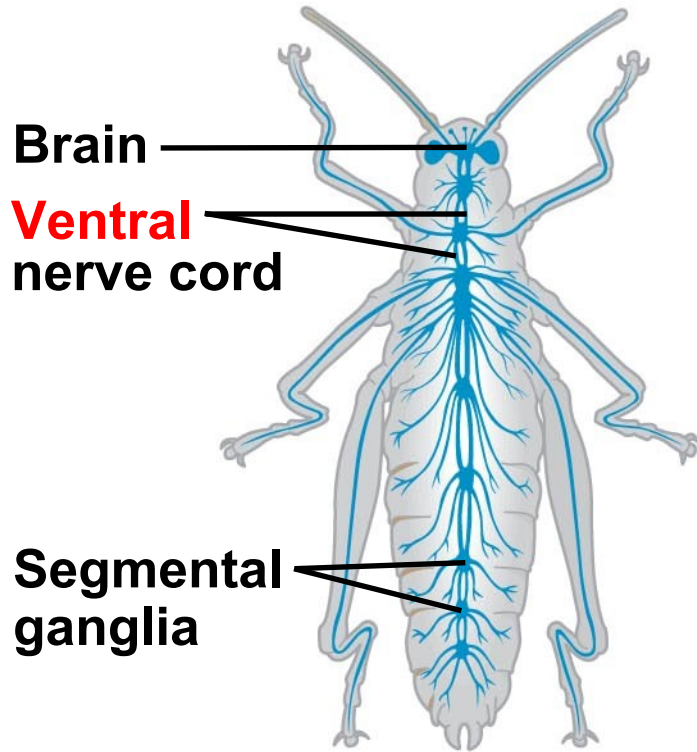


Planarian (flatworm)

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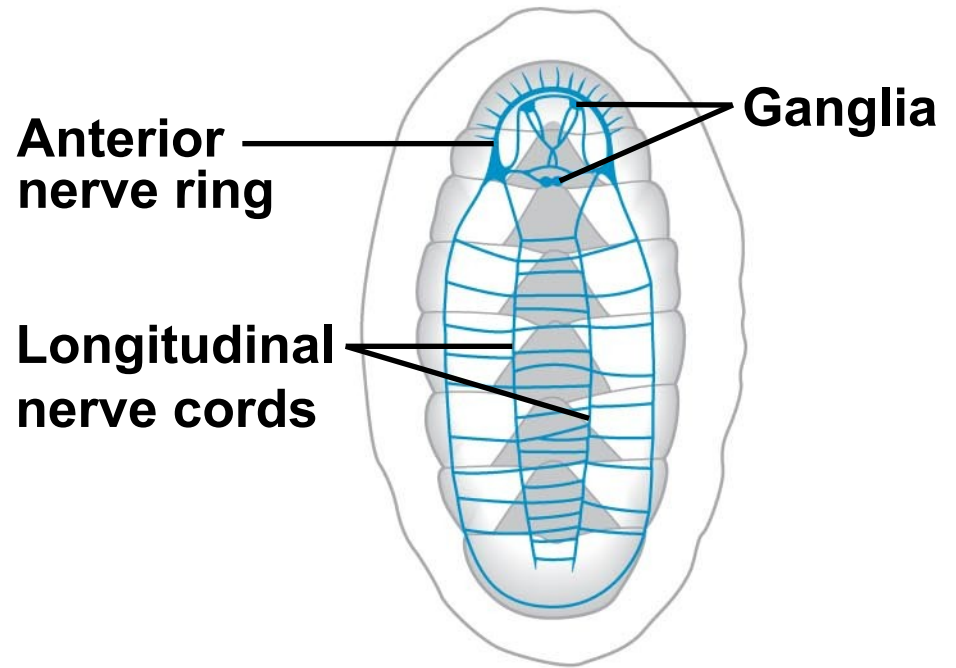


Leech (annelid)



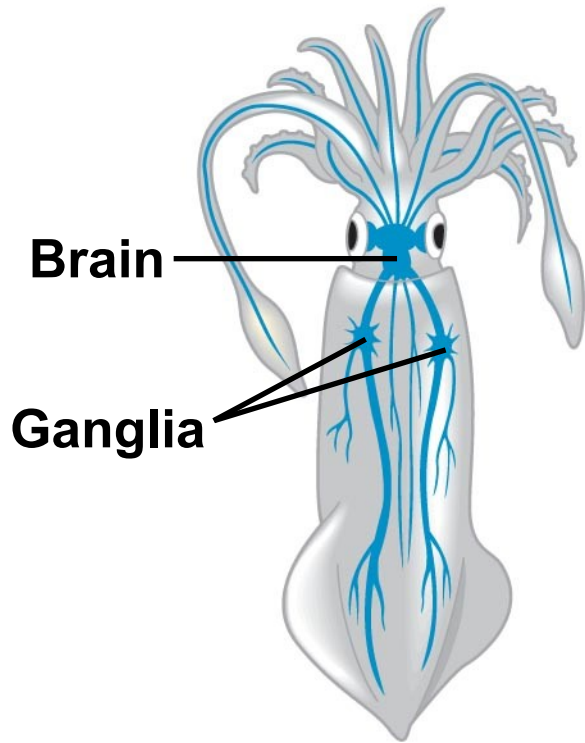
Insect (arthropod)

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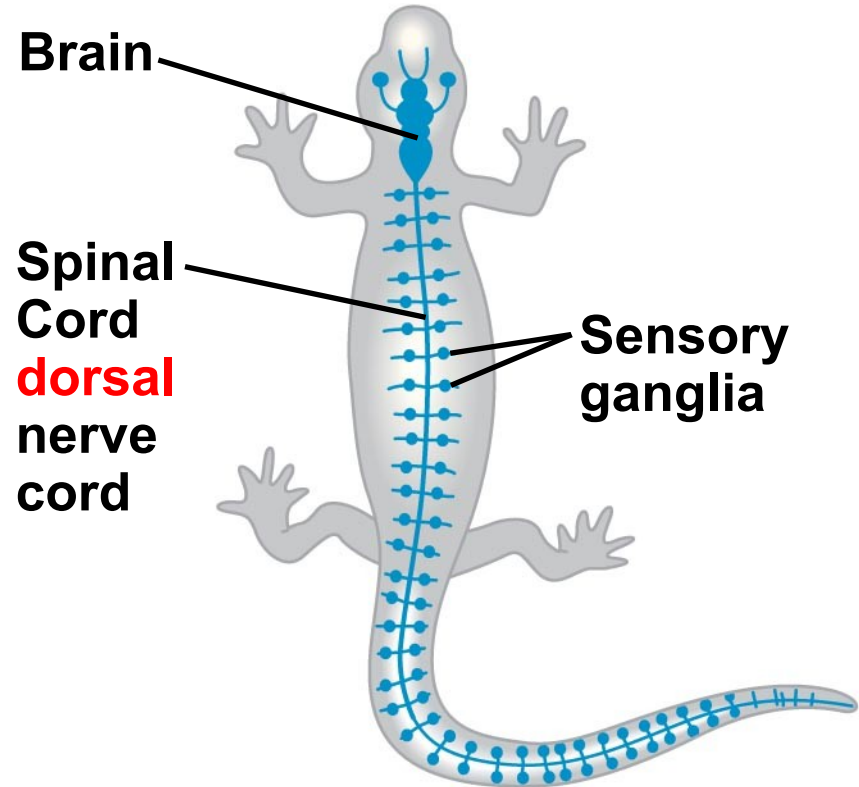
Chiton (mollusc)

-
- Annelids and arthropods have segmentally arranged clusters of neurons called ganglia.
 - *Nervous system organization usually correlates with lifestyle.*
 - Sessile molluscs (e.g., clams and chitons) have simple systems, whereas more complex molluscs (e.g., octopuses and squids) have more sophisticated systems.



Squid (mollusc)

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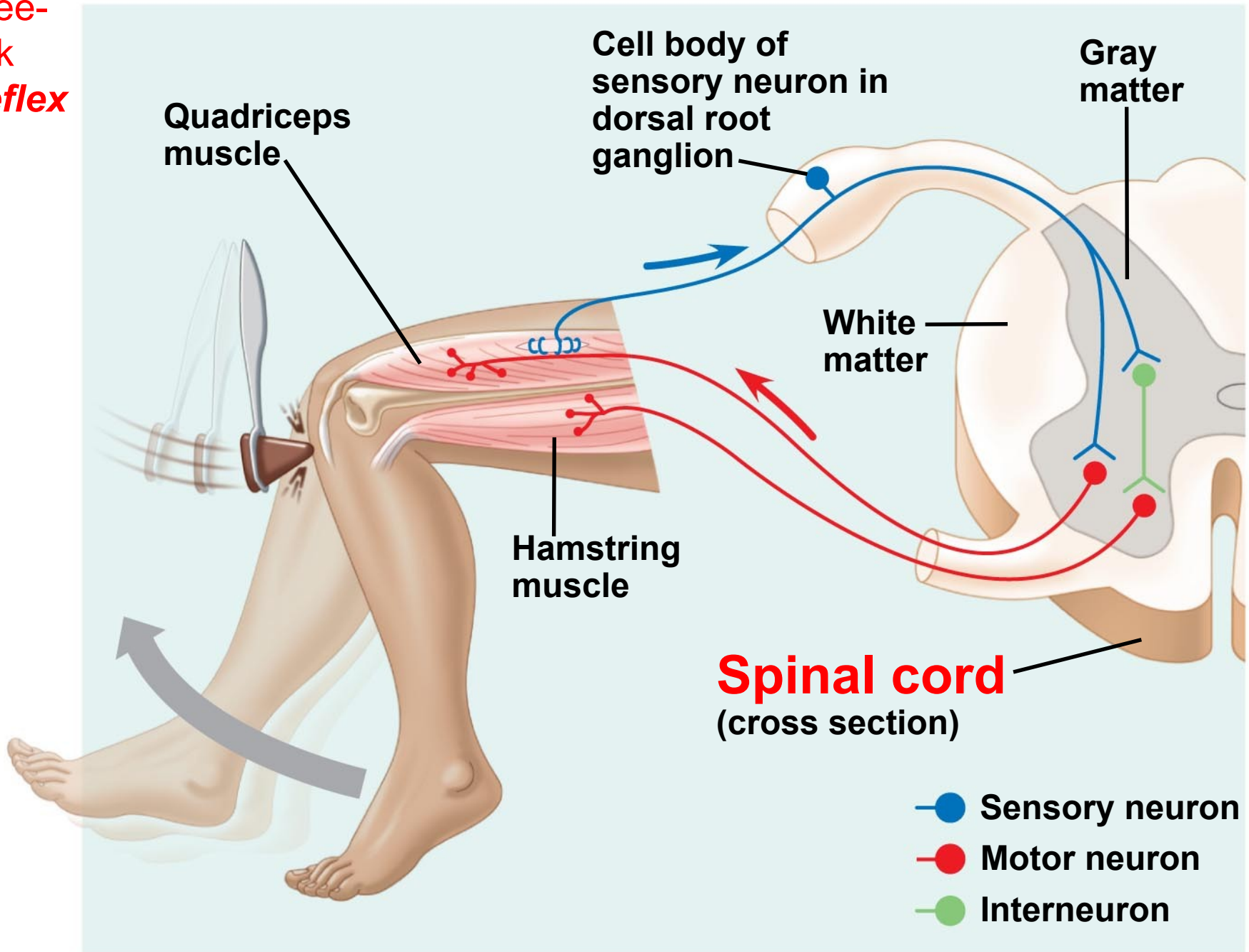
Salamander (vertebrate)

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- In vertebrates
 - The **CNS** is composed of the brain and spinal cord.
 - The *peripheral nervous system* (**PNS**) is composed of nerves and ganglia.

Organization of the Vertebrate Nervous System

- The spinal cord conveys information from the brain to the PNS.
- The **spinal cord** also produces reflexes independently of the brain.
- A **reflex** is the body's **automatic response to a stimulus**.
 - For example, a doctor uses a mallet to trigger a knee-jerk reflex.

knee-jerk
Reflex

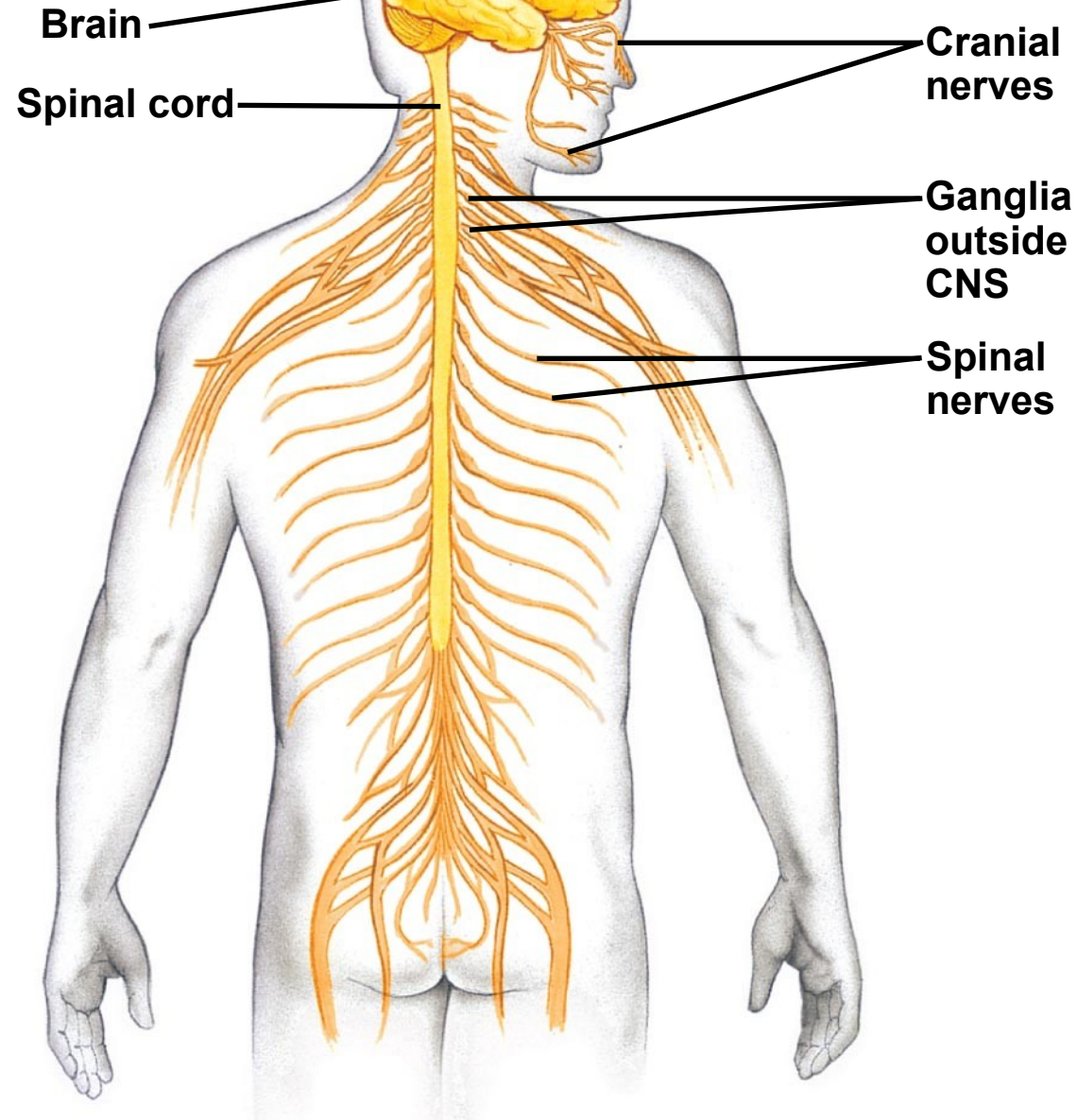


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- **Invertebrates** usually have a **ventral** nerve cord while **vertebrates** have a **dorsal** spinal cord.
 - The spinal cord and brain develop from the embryonic nerve cord.

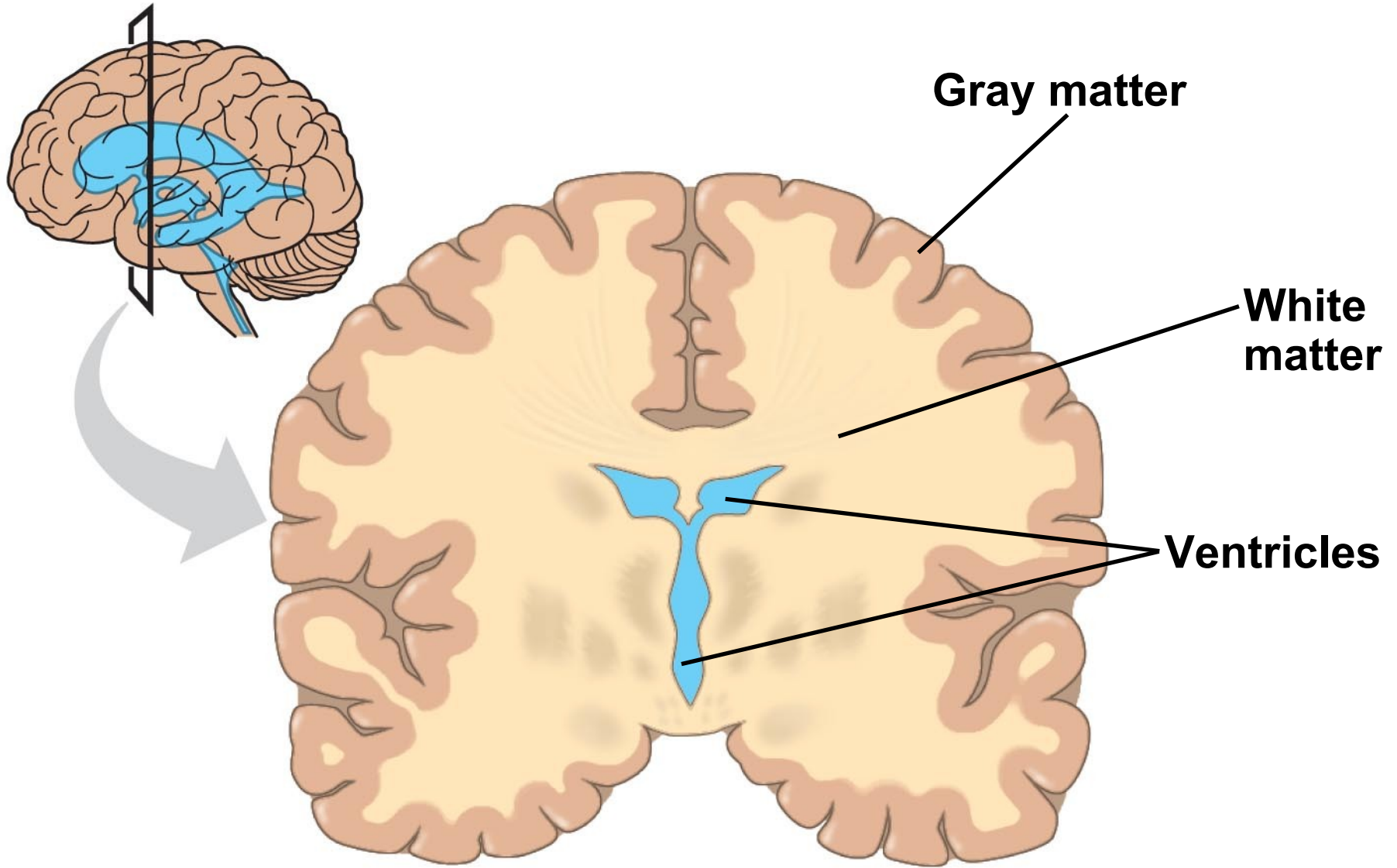
Vertebrate Nervous System

**Central nervous
system (CNS)**

**Peripheral nervous
system (PNS)**



Ventricles, gray matter, and white matter



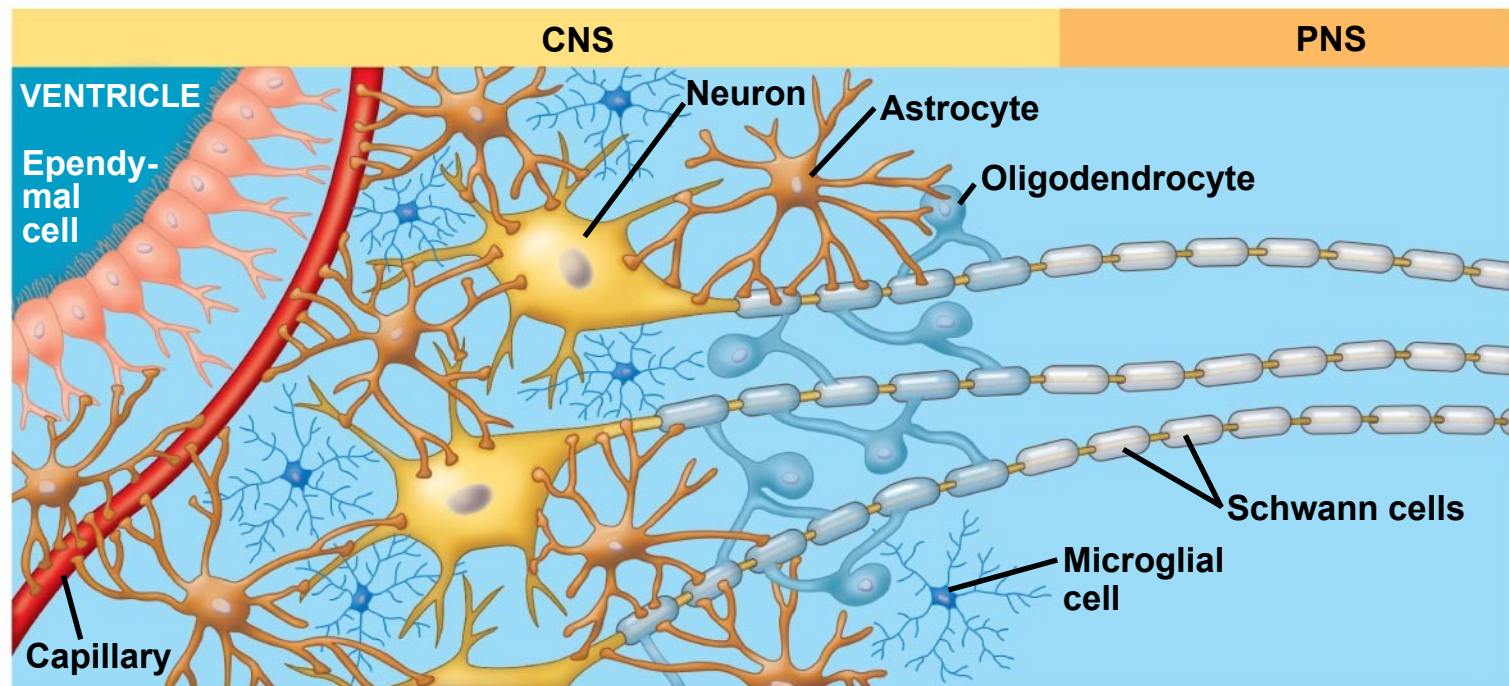
-
- The **central canal** of the **spinal cord** and the **ventricles** of the **brain** are hollow and filled with **cerebrospinal fluid**.
 - The cerebrospinal fluid is filtered from blood and functions to cushion the brain and spinal cord.

-
- The **brain** and **spinal cord** contain
 - **Gray matter**, which consists of neuron cell bodies, dendrites, and unmyelinated axons.
 - **White matter**, which consists of bundles of **myelinated axons**.

Glia in the CNS

- Glia have numerous functions
 - Ependymal cells promote **circulation** of **cerebrospinal fluid**.
 - Microglia **protect** the nervous system from **microorganisms**.
 - Oligodendrocytes and Schwann cells form the **myelin sheaths** around axons.

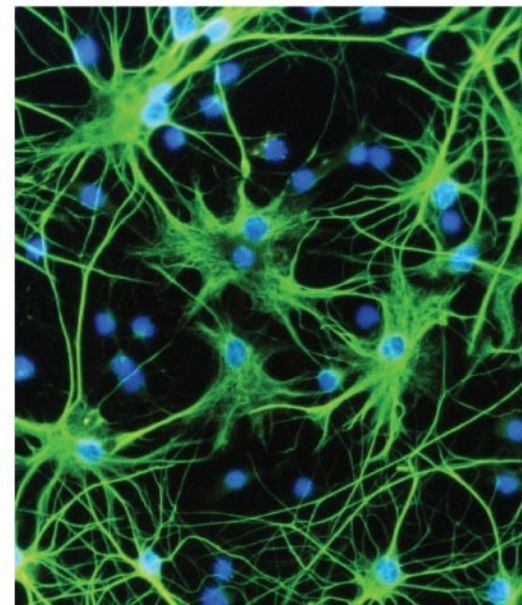
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- Glia have numerous functions
 - **Astrocytes** provide *structural support* for neurons, regulate extracellular ions and neurotransmitters, and induce the formation of a *blood-brain barrier* that regulates the *chemical environment* of the CNS
 - **Radial glia** play a role in the embryonic development of the nervous system.



(a) Glia in vertebrates

Glia in the vertebrate nervous system

50 μm

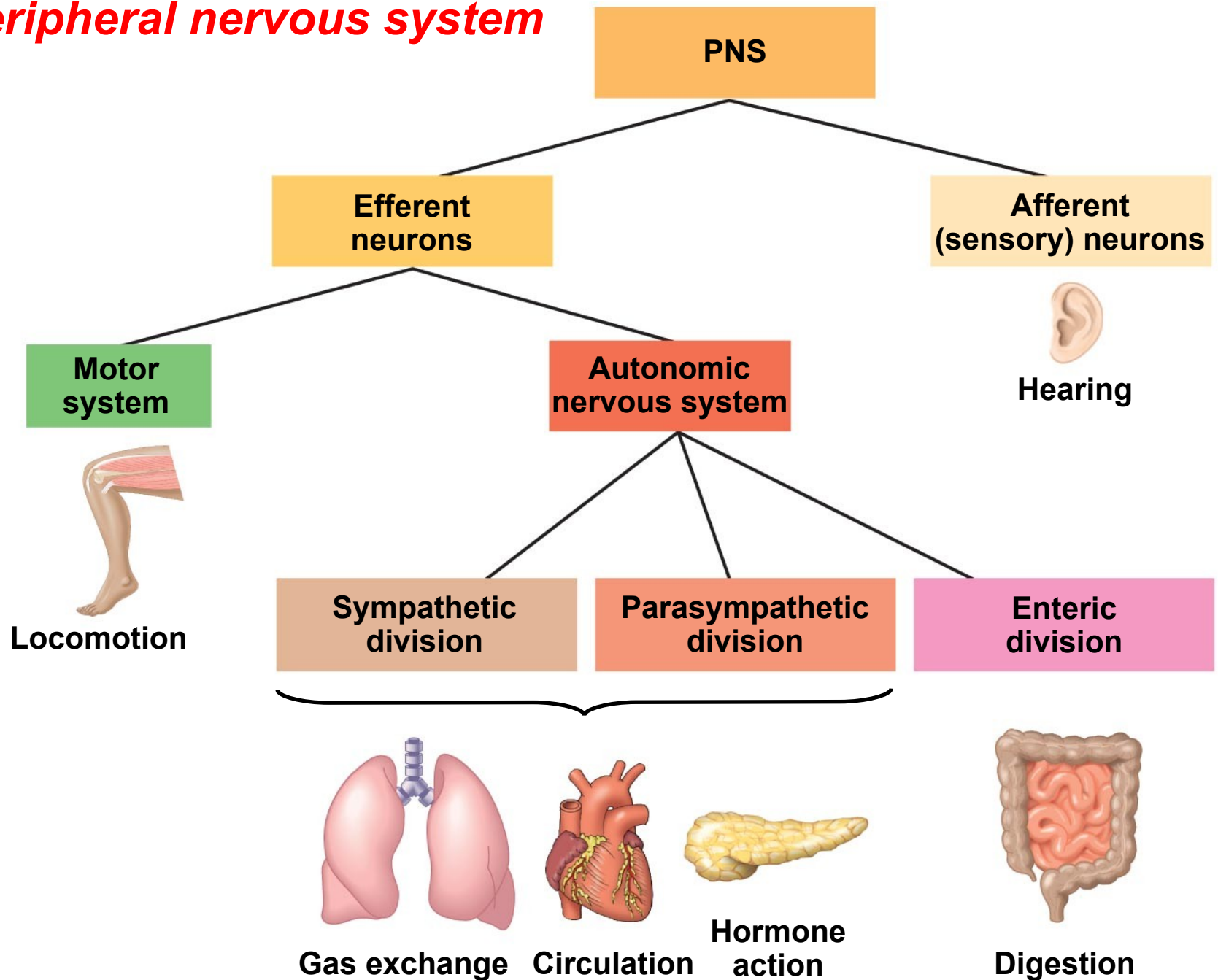


(b) Astrocytes (LM)

The Peripheral Nervous System

- The PNS transmits information to and from the CNS and regulates movement and the internal environment.
- In the **PNS**, *afferent neurons* transmit information to the CNS and *efferent neurons* transmit information away from the CNS.
- **Cranial nerves** originate in the brain and mostly terminate in organs of the head and upper body.
- **Spinal nerves** originate in the spinal cord and extend to parts of the body below the head.

peripheral nervous system



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- The **PNS** has two functional components: the motor system and the autonomic nervous system.
 - The **motor system** carries signals to skeletal muscles and is voluntary.
 - The **autonomic nervous system** regulates the internal environment in an involuntary manner.

-
- The PNS **autonomic nervous system** has sympathetic, parasympathetic, and enteric divisions
 - The **sympathetic** and **parasympathetic** divisions have **antagonistic effects** on target organs.

-
- The **sympathetic division** correlates with the “fight-or-flight” response.
 - The **parasympathetic division** promotes a return to “rest and digest.”
 - The **enteric division** controls activity of the digestive tract, pancreas, and gallbladder.

PNS:
autonomic
nervous
system

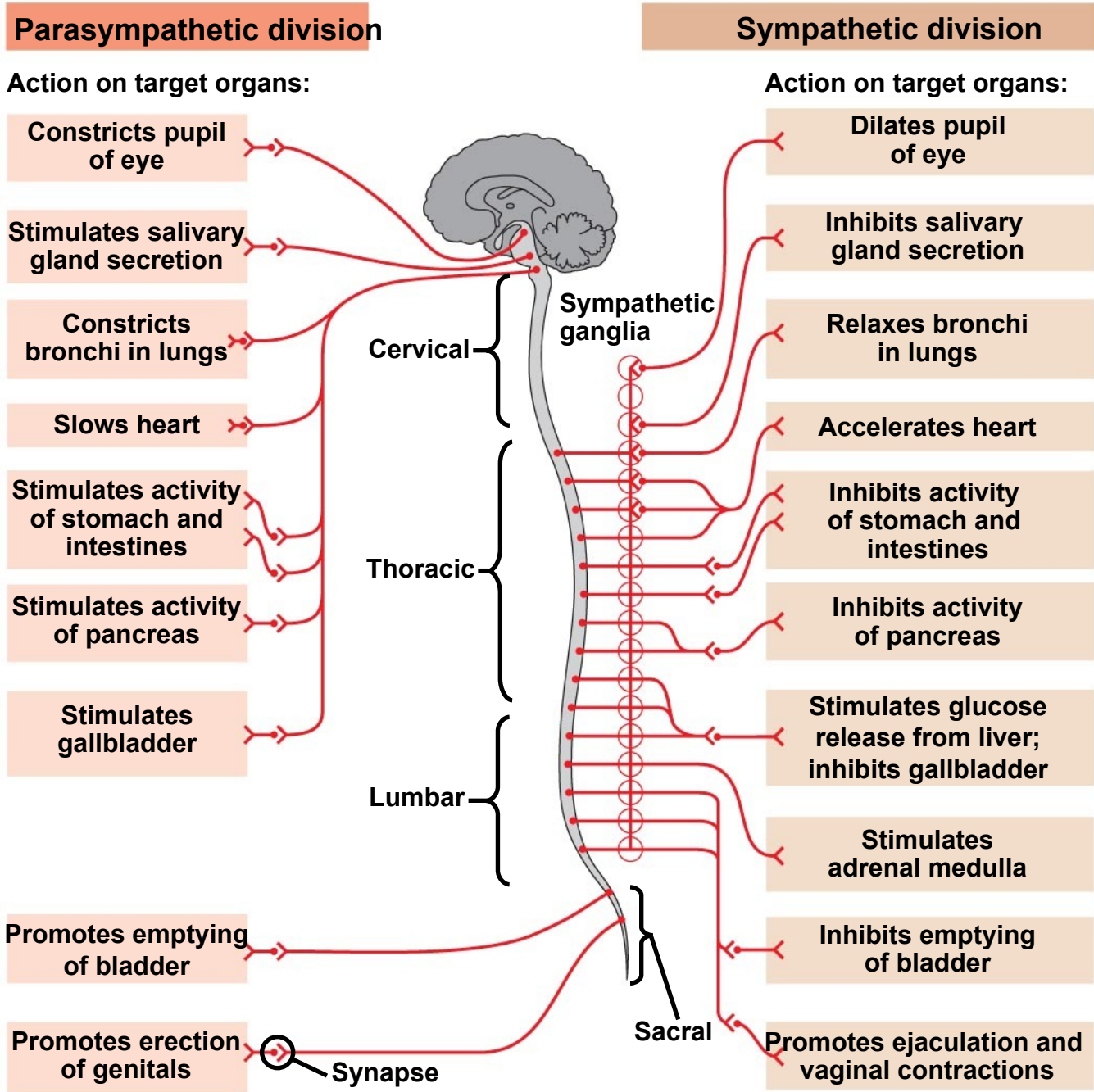


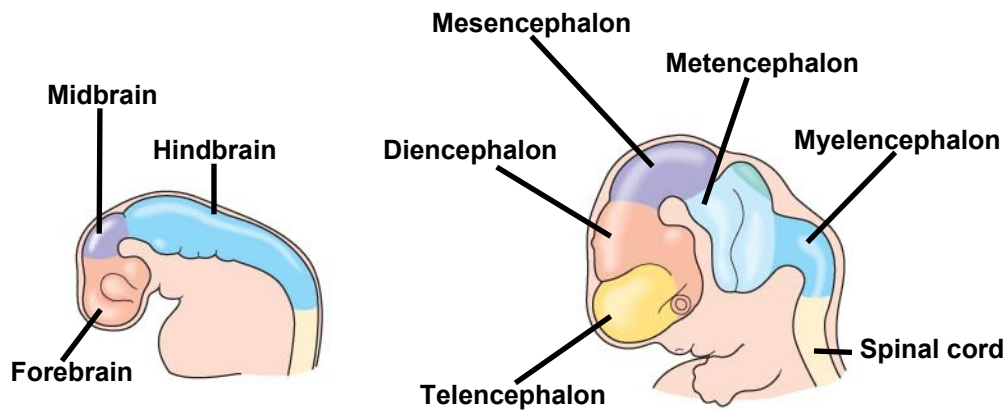
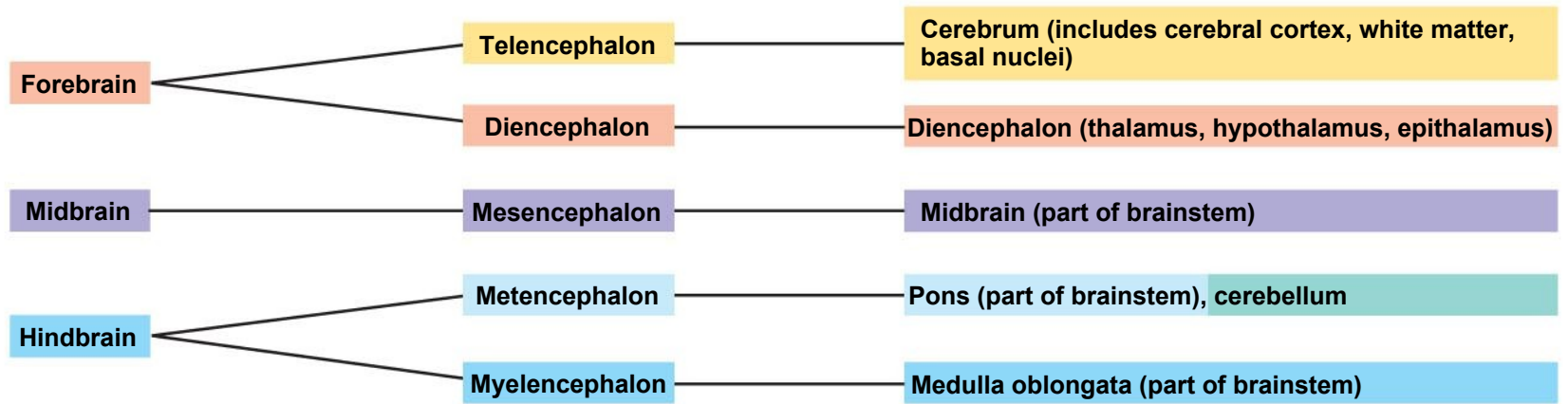
Table 49.1 Properties of Parasympathetic and Sympathetic Neurons

	Parasympathetic Division	Sympathetic Division
Preganglionic Neurons		
Location	Brainstem, sacral segments of spinal cord	Thoracic and lumbar segments of spinal cord
Neurotransmitter released	Acetylcholine	Acetylcholine
Postganglionic Neurons		
Location	Ganglia close to or within target organs	Ganglia close to target organs or chain of ganglia near spinal cord
Neurotransmitter released	Acetylcholine	Norepinephrine

The vertebrate brain is regionally specialized

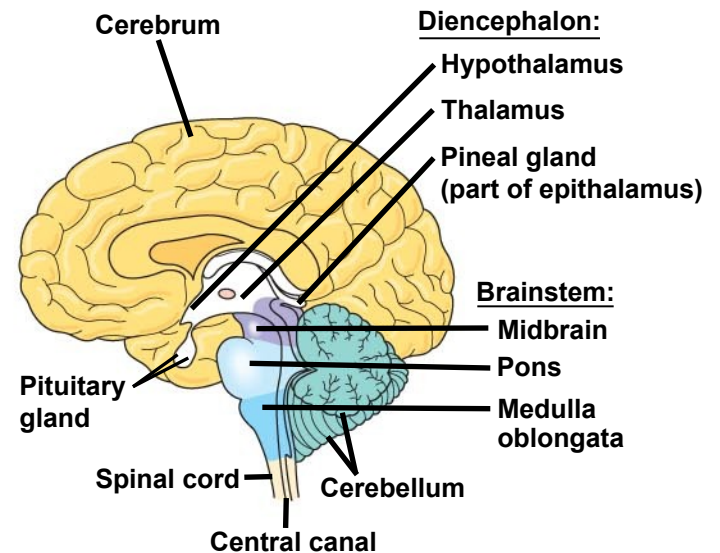
- All vertebrate brains develop from three embryonic regions: **forebrain**, **midbrain**, and **hindbrain**.
- By the fifth week of human embryonic development, five brain regions have formed from the three embryonic regions.

Development of the human brain



(a) Embryo at 1 month

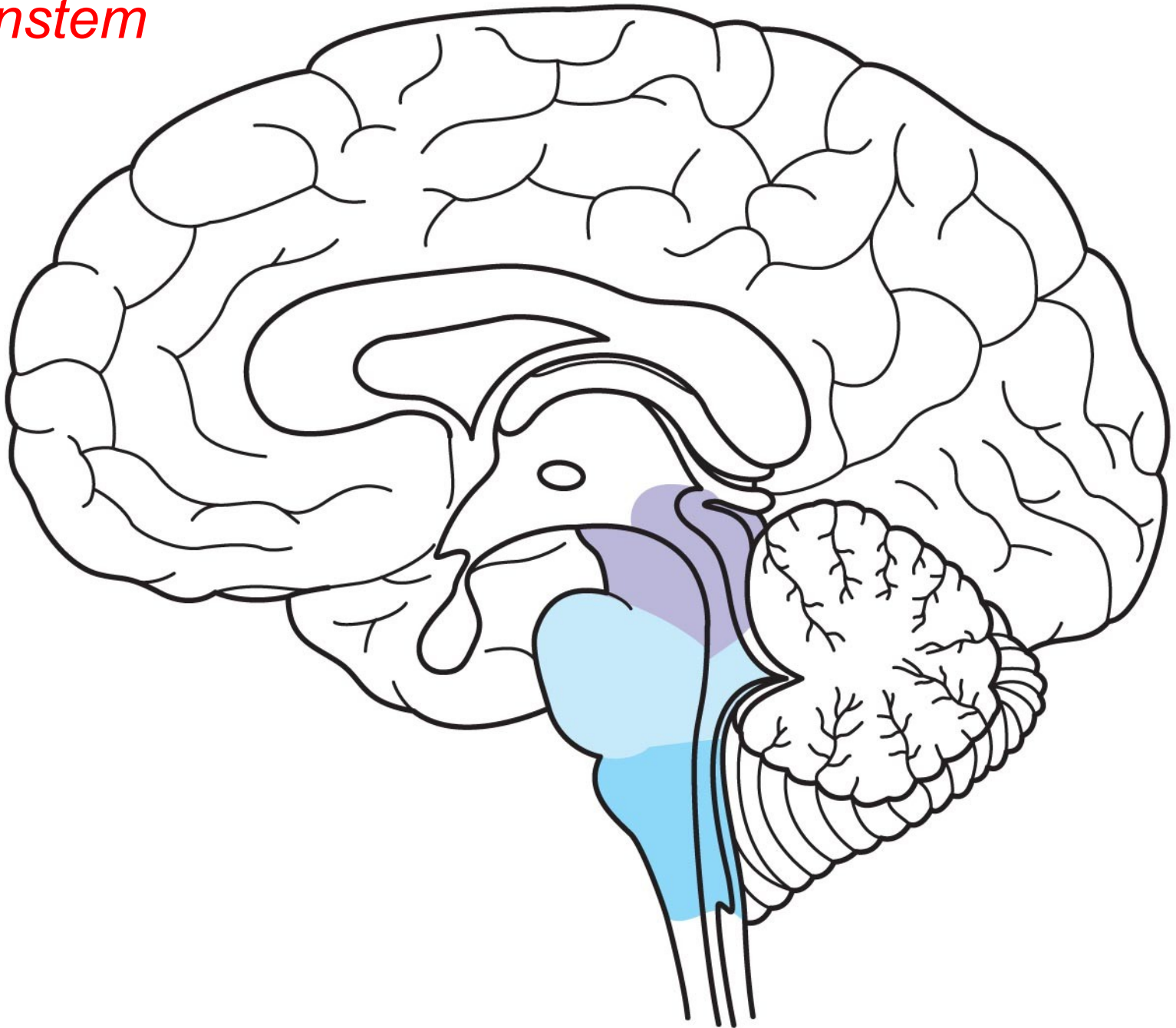
(b) Embryo at 5 weeks



(c) Adult

-
- As a human brain develops further, the most profound change occurs in the forebrain, which gives rise to the **cerebrum**.
 - The outer portion of the cerebrum called the **cerebral cortex** surrounds much of the brain.

Brainstem



The Brainstem

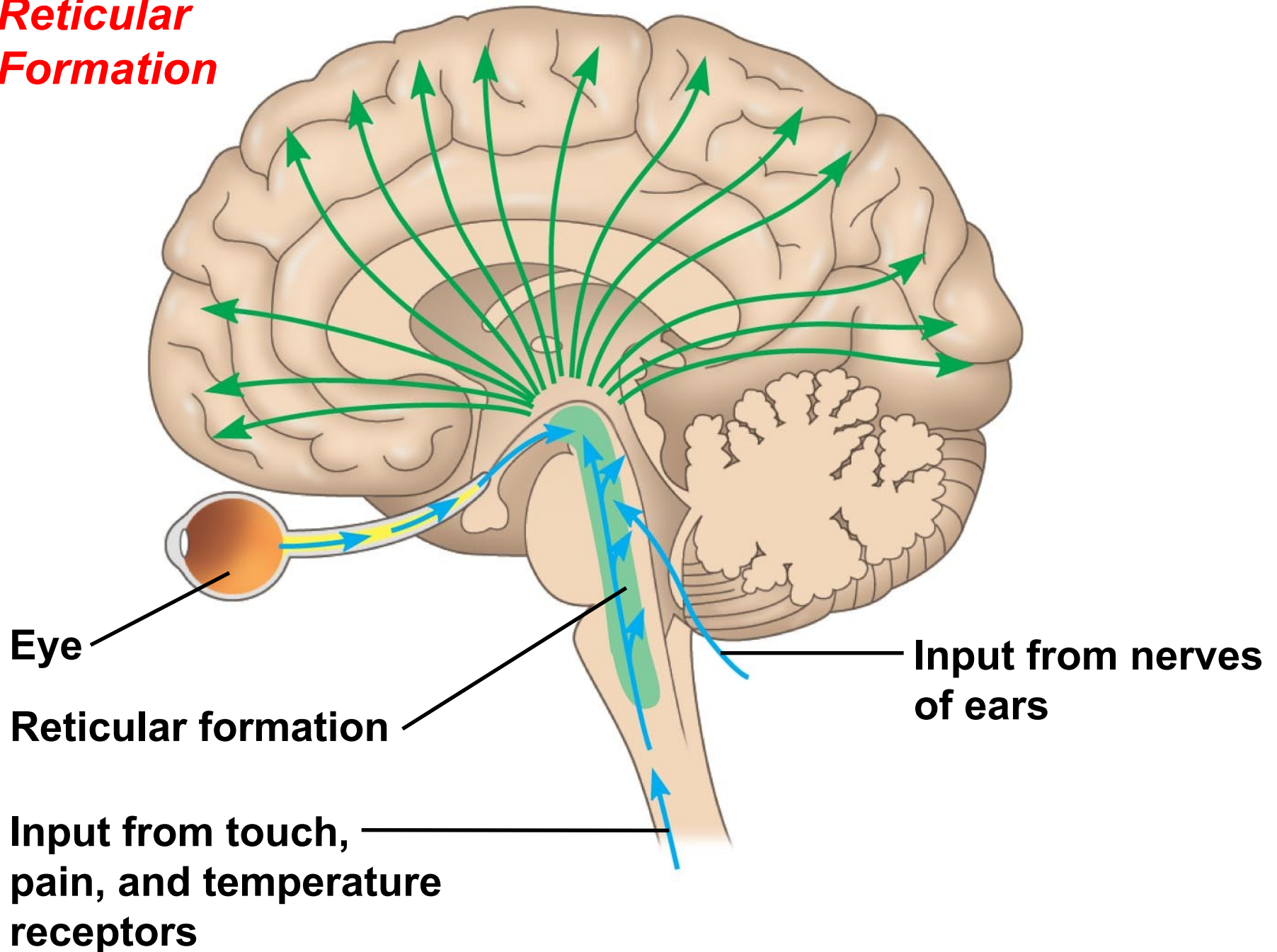
- The **brainstem** coordinates and conducts **information** between brain centers.
- The brainstem has **three parts**: the **midbrain**, the **pons**, and the **medulla oblongata**.

-
- The midbrain contains centers for receipt and integration of sensory information.
 - The pons regulates breathing centers in the medulla.
 - The medulla oblongata contains centers that control several functions including breathing, cardiovascular activity, swallowing, vomiting, and digestion.

Arousal and Sleep

- The **brainstem** and **cerebrum** control arousal and sleep.
- The core of the brainstem has a diffuse network of neurons called the **reticular formation**.
- This regulates the amount and type of information that reaches the cerebral cortex and affects **alertness**.
- The hormone **melatonin** is released by the **pineal gland** and plays a role in bird and mammal **sleep cycles**.

Reticular Formation

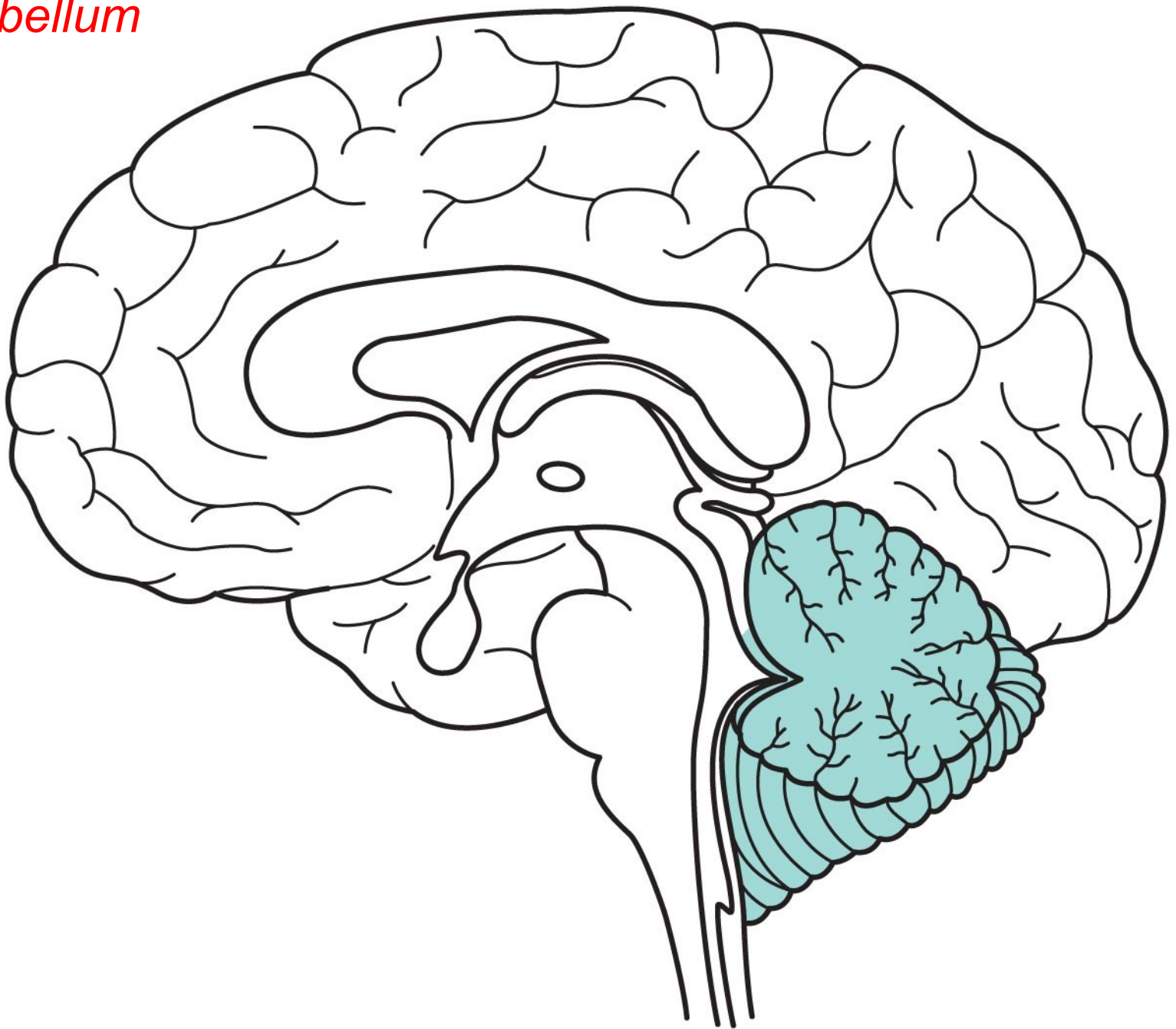


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- Sleep is essential and may play a role in the consolidation of learning and memory.
 - Dolphins sleep with one brain hemisphere at a time and are therefore able to swim while “asleep.”

The Cerebellum

- The **cerebellum** is important for **coordination** and error checking during motor, perceptual, and cognitive functions.
- It is also involved in learning and remembering **motor skills**.

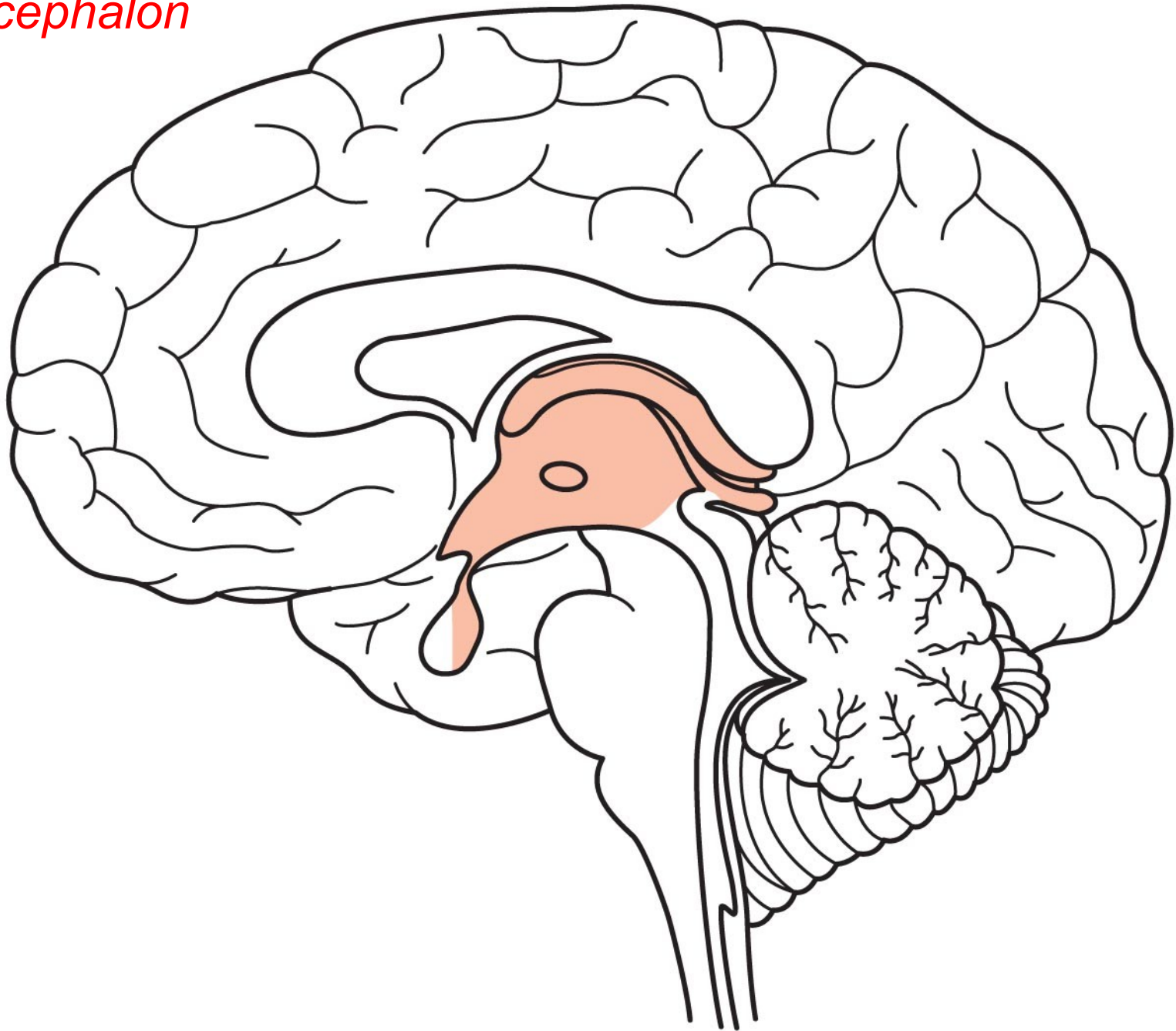
Cerebellum



The Diencephalon

- The diencephalon develops into **three regions**: the epithalamus, thalamus, and hypothalamus.
- The **epithalamus** includes the pineal gland and generates cerebrospinal fluid from blood.
- The **thalamus** is the main input center for sensory information to the cerebrum and the main output center for motor information leaving the cerebrum.
- The **hypothalamus** regulates homeostasis and basic survival behaviors such as feeding, fighting, fleeing, and reproducing.

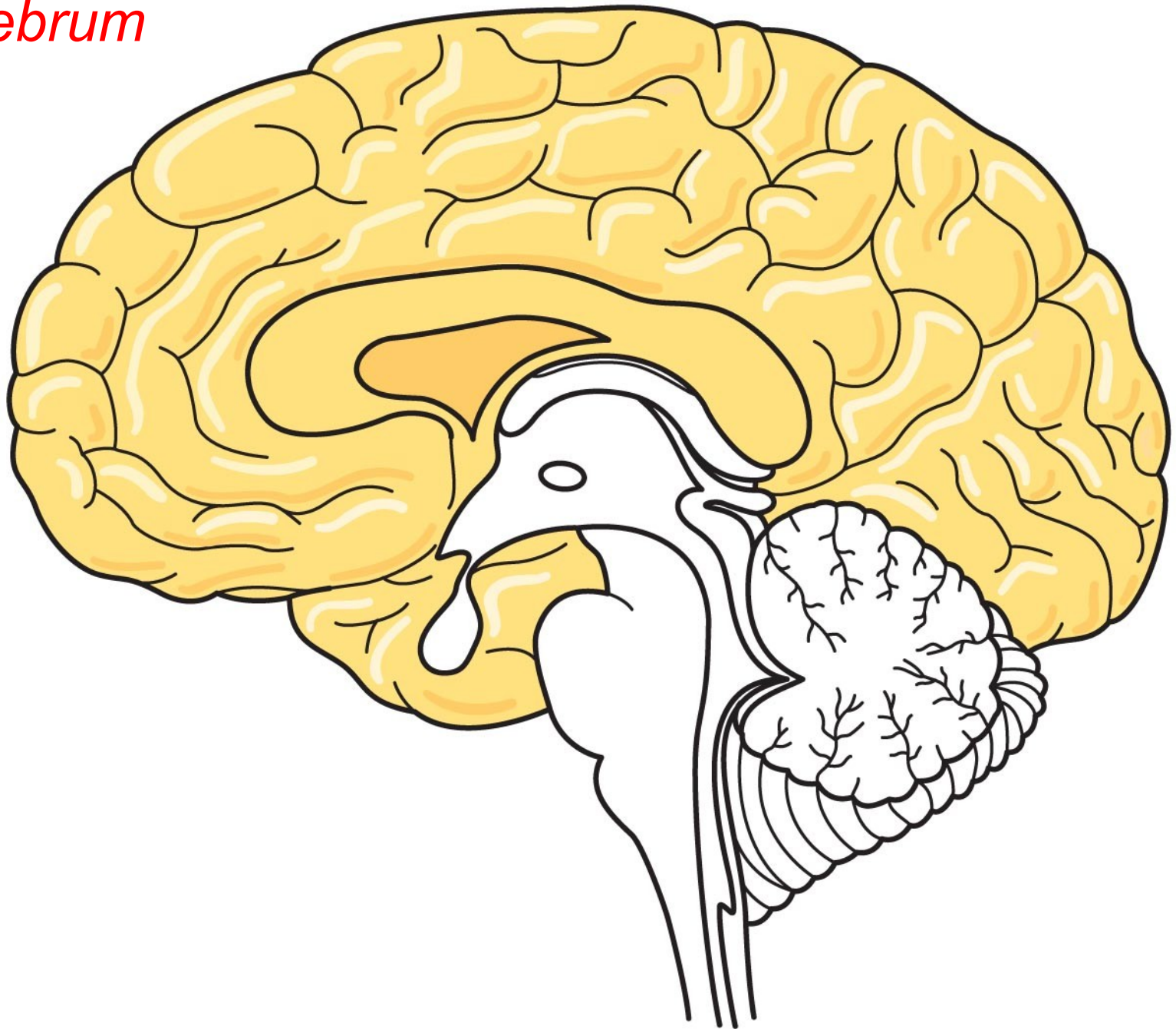
Diencephalon



Biological Clock Regulation by the Hypothalamus

- The hypothalamus also regulates **circadian rhythms** such as the **sleep/wake cycle**.
- Mammals usually have a pair of **suprachiasmatic nuclei (SCN)** in the hypothalamus that function as a **biological clock**.
- Biological clocks usually require external cues to remain synchronized with environmental cycles.

Cerebrum



Cerebrum

- The cerebrum has right and left **cerebral hemispheres**.
- Each cerebral hemisphere consists of a cerebral cortex (gray matter) overlying white matter and basal nuclei.
- In humans, the **cerebral cortex** is the **largest** and most complex part of the brain.
- The basal nuclei are important centers for planning and **learning** movement sequences.

-
- A thick band of axons called the **corpus callosum** provides communication between the right and left cerebral cortices.
 - The **right** half of the cerebral cortex controls the **left** side of the body, and vice versa.

Human Brain viewed from the rear

Left cerebral hemisphere

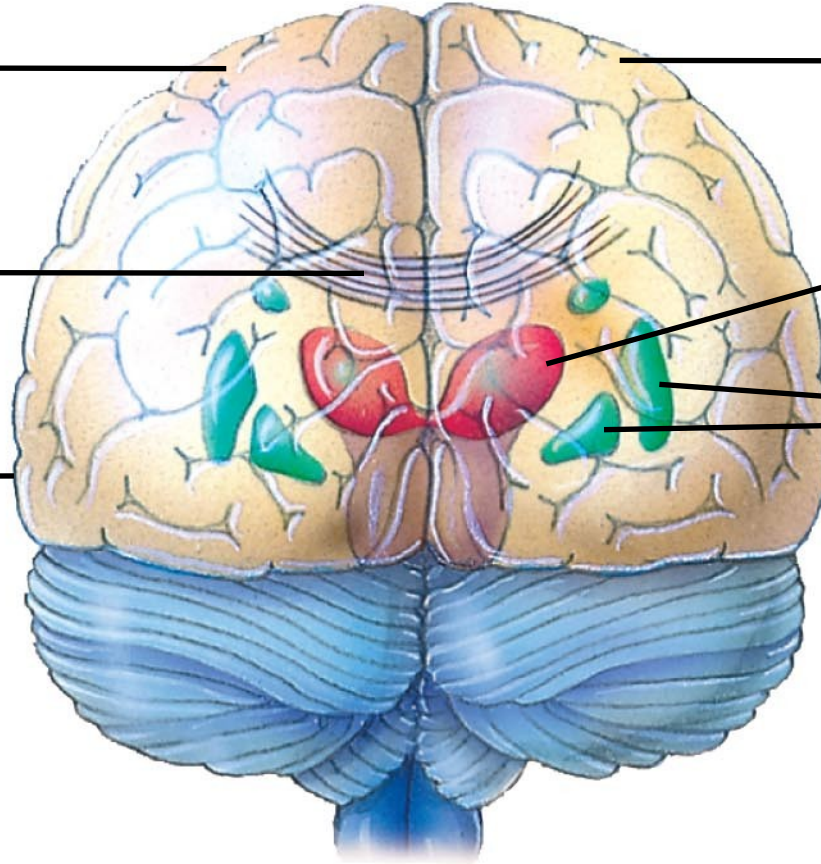
Right cerebral hemisphere

Corpus callosum

Thalamus

Cerebral cortex

Basal nuclei



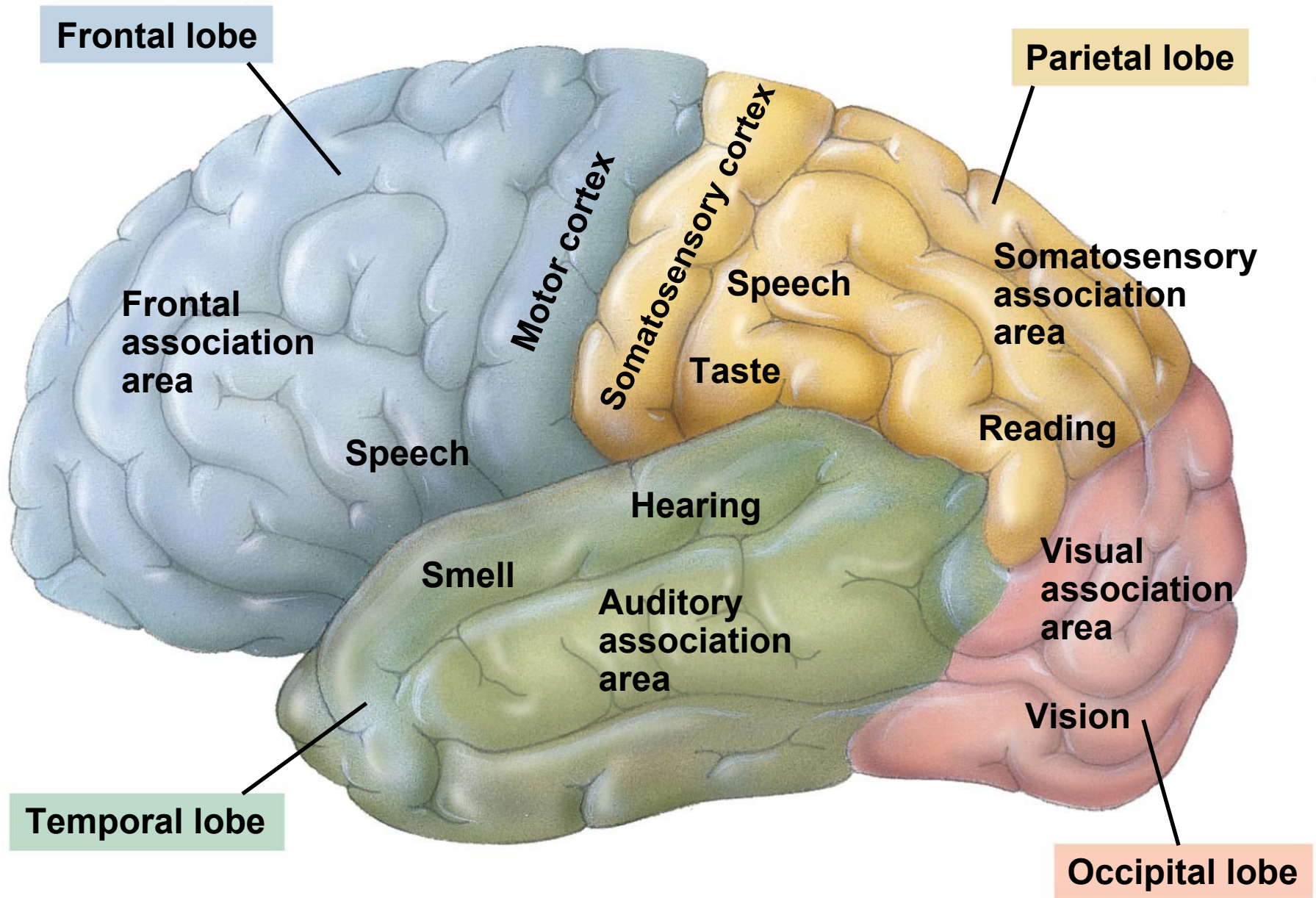
Evolution of Cognition in Vertebrates

- The outermost layer of the cerebral cortex has a different arrangement in birds and mammals.
- In mammals, the cerebral cortex has a convoluted surface called the *neocortex*, which was previously thought to be required for cognition.
- *Cognition is the perception and reasoning that form knowledge.*
- However, it has recently been shown that birds also demonstrate cognition even though they lack a neocortex.

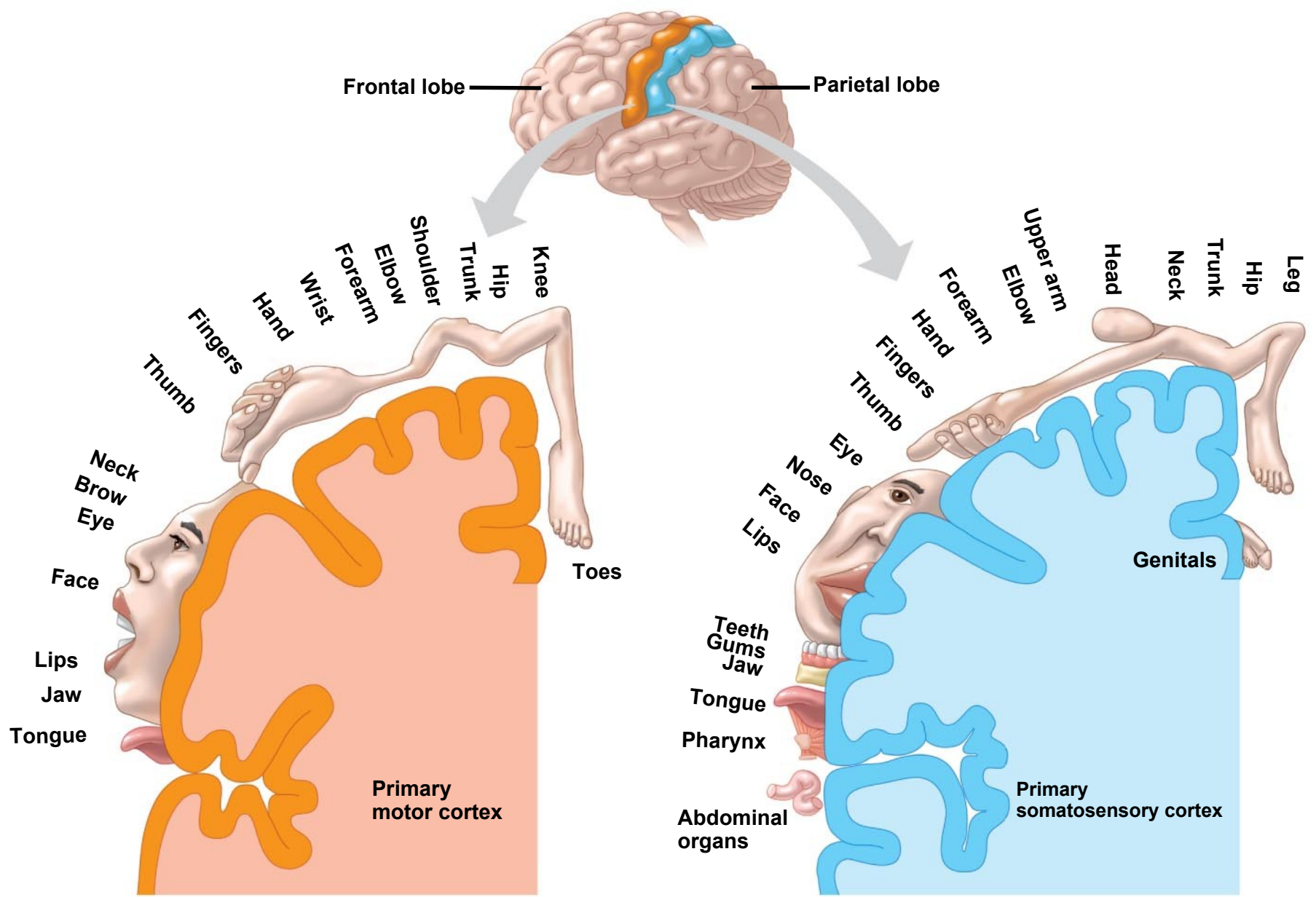
The cerebral cortex controls voluntary movement and cognitive functions

- Each side of the cerebral cortex has four lobes: frontal, temporal, occipital, and parietal.
- Each lobe contains primary sensory areas and association areas where information is integrated.

human cerebral cortex



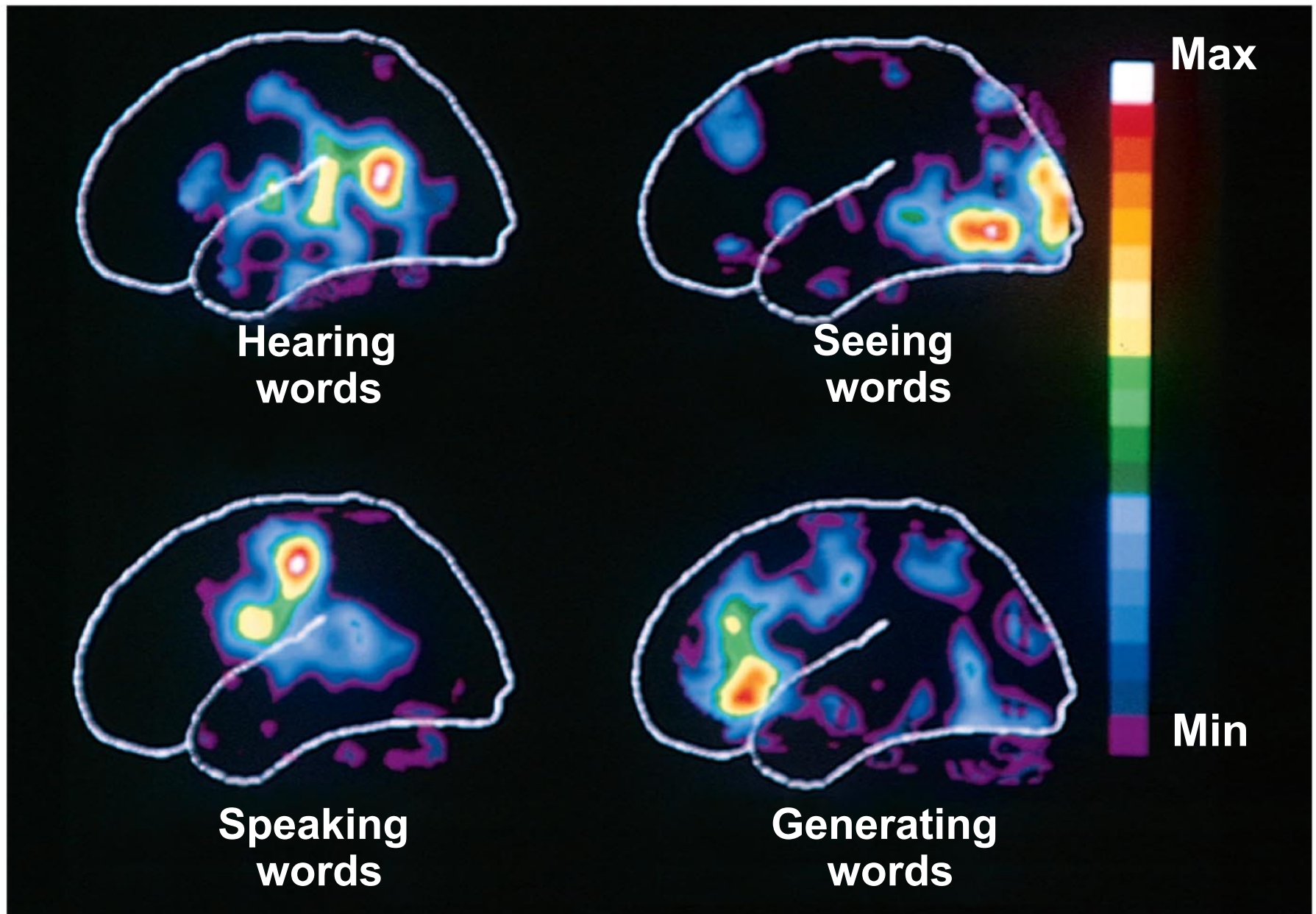
Body part representation in primary motor and primary somatosensory cortices



Language and Speech

- Studies of brain activity have mapped areas responsible for language and speech.
- **Broca's area** in the frontal lobe is active when **speech** is generated.
- Wernicke's area in the temporal lobe is active when speech is heard.

Mapping language areas in the cerebral cortex



Lateralization of Cortical Function

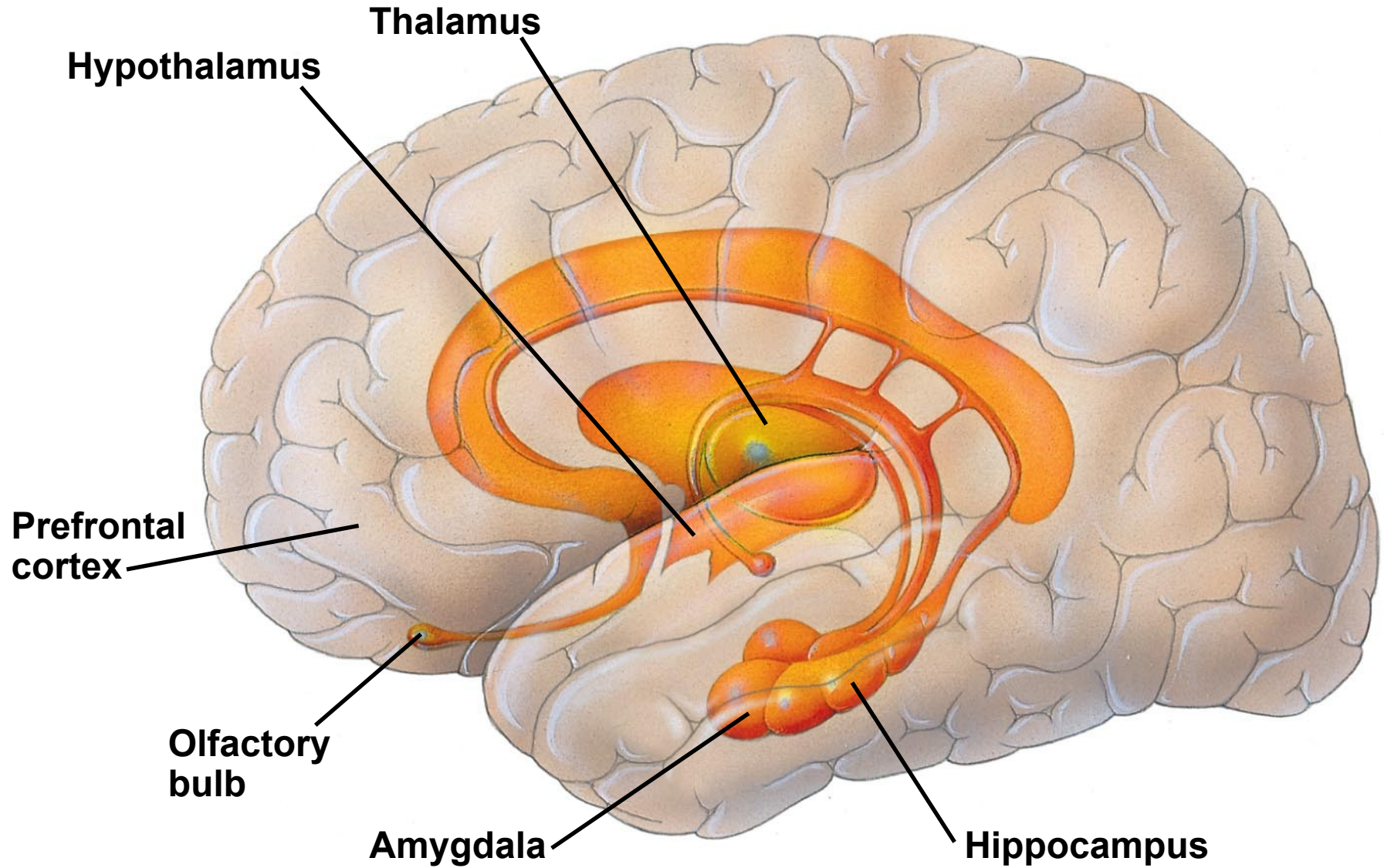
- The **corpus callosum** transmits information between the two cerebral hemispheres.
- The **left hemisphere** is more adept at language, math, logic, and processing of serial sequences.
- The **right hemisphere** is stronger at pattern recognition, nonverbal thinking, and emotional processing.

-
- The differences in hemisphere function are called **lateralization**.
 - Lateralization is linked to handedness.

Emotions

- Emotions are generated and experienced by the ***limbic system*** and other parts of the brain including the sensory areas.
- The *limbic system* is a ring of structures around the ***brainstem*** that includes the **amygdala**, **hippocampus**, and parts of the **thalamus**.
- The **amygdala** is located in the temporal lobe and helps store an emotional experience as an **emotional memory**.

The limbic system



Neural Plasticity

- **Neural plasticity** describes the **ability** of the nervous system to be **modified** after birth.
- Changes can strengthen or weaken signaling at a synapse.

Memory and Learning

- Learning can occur when neurons make new connections or when the strength of existing neural connections changes.
- **Short-term memory** is accessed via the hippocampus.
- The hippocampus also plays a role in forming **long-term memory**, which is stored in the cerebral cortex.

Nervous system disorders can be explained in molecular terms

- Disorders of the nervous system include schizophrenia, depression, Alzheimer's disease, and Parkinson's disease.
- Genetic and environmental factors contribute to diseases of the nervous system.

Schizophrenia

- About 1% of the world's population suffers from **schizophrenia**.
- Schizophrenia is characterized by hallucinations, delusions, blunted emotions, and other symptoms.
- Available treatments focus on brain pathways that use dopamine as a neurotransmitter.

Depression

- Two broad forms of depressive illness are known: major depressive disorder and bipolar disorder.
- In **major depressive disorder**, patients have a persistent lack of interest or pleasure in most activities.
- **Bipolar disorder** is characterized by manic (high-mood) and depressive (low-mood) phases.
- Treatments for these types of depression include drugs such as Prozac and lithium.

Drug Addiction and the *Brain Reward System*

- The brain's reward system rewards motivation with pleasure.
- Some drugs are addictive because they increase activity of the brain's reward system.
- These drugs include cocaine, amphetamine, heroin, alcohol, and tobacco.
- Drug addiction is characterized by compulsive consumption and an inability to control intake.

-
- Addictive drugs enhance the activity of the dopamine pathway.
 - Drug addiction leads to long-lasting changes in the reward circuitry that cause craving for the drug.

Effects of addictive drugs on the reward pathway of the mammalian brain

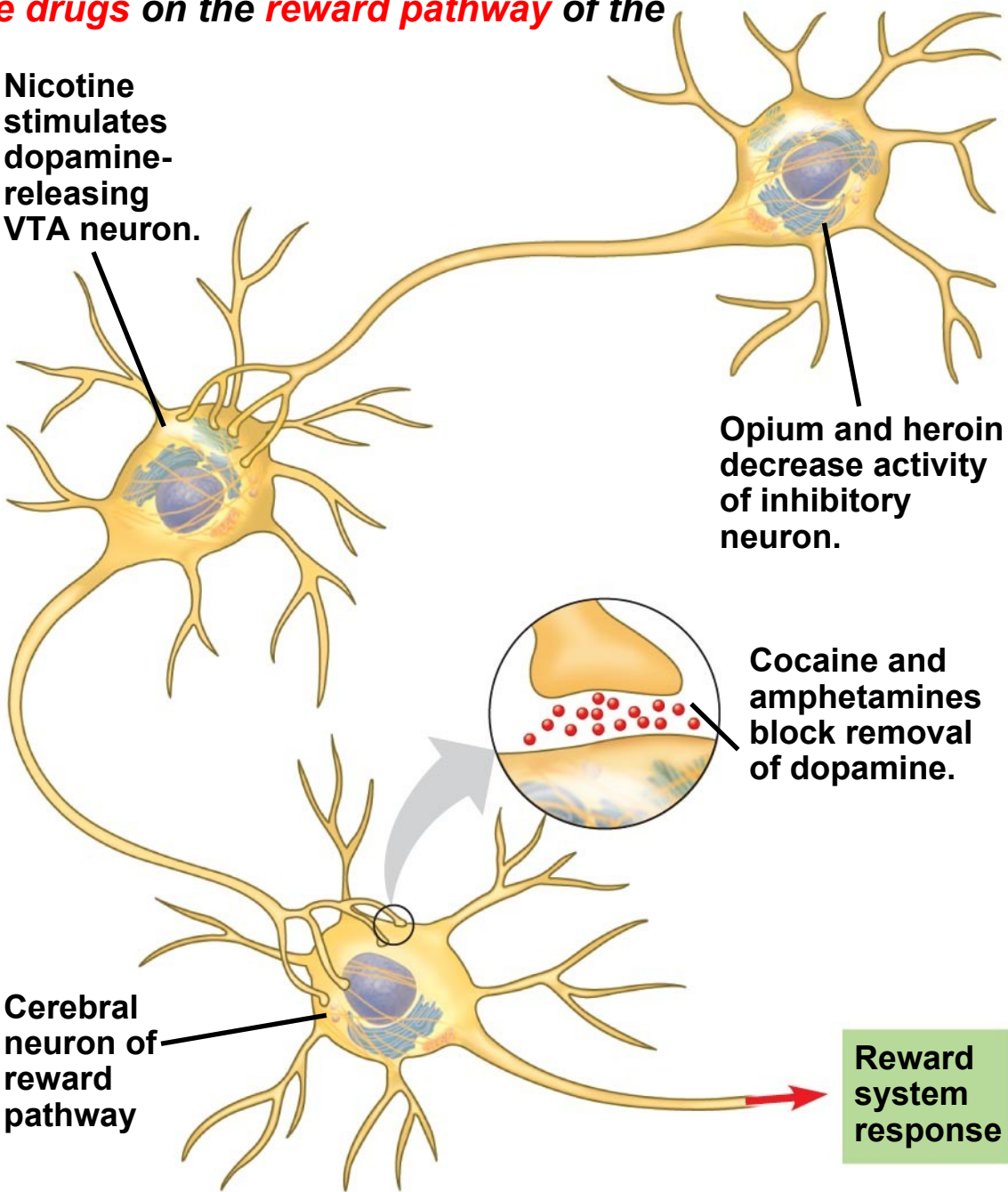
Nicotine stimulates dopamine-releasing VTA neuron.

Opium and heroin decrease activity of inhibitory neuron.

Cocaine and amphetamines block removal of dopamine.

Cerebral neuron of reward pathway

Reward system response



Alzheimer's Disease

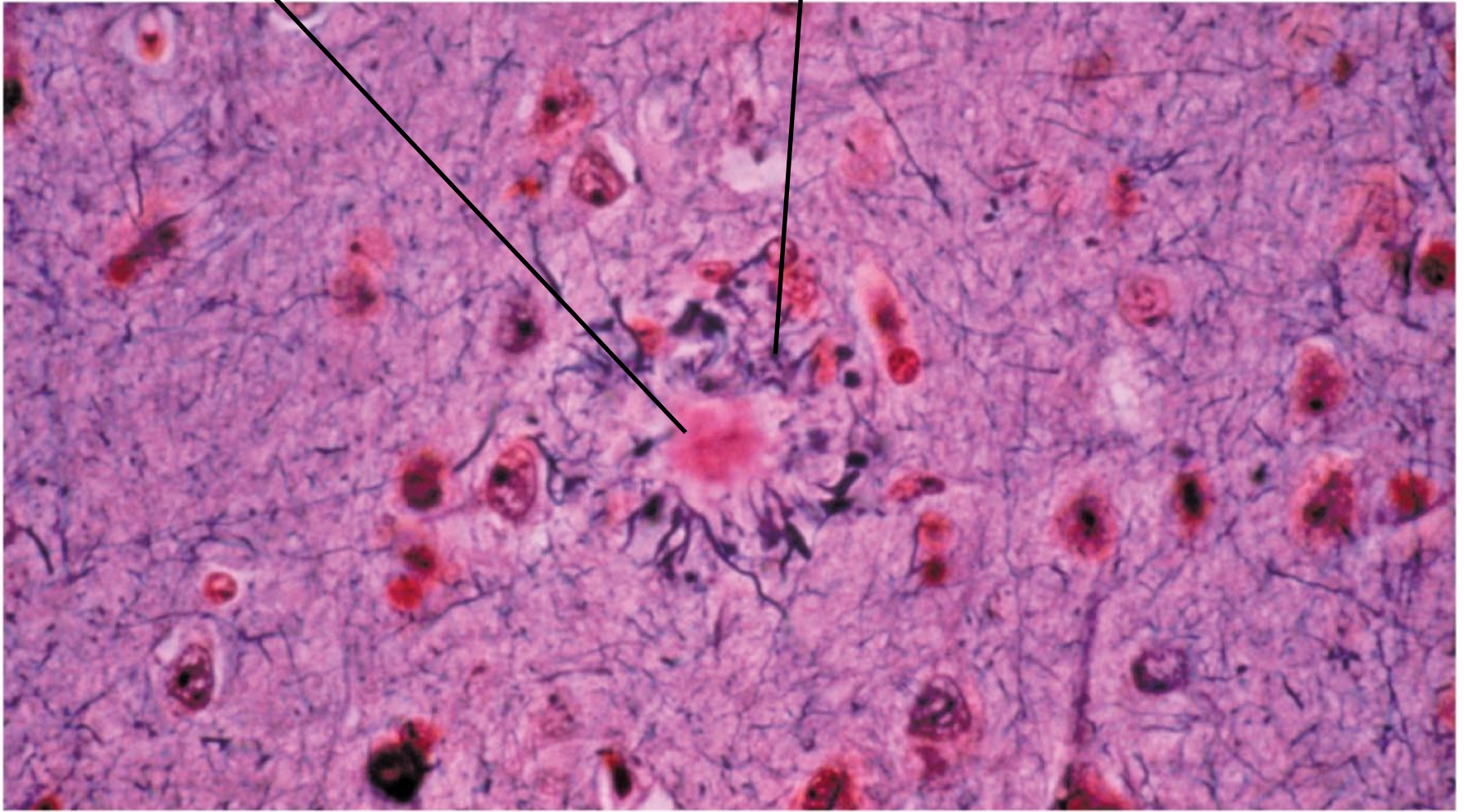
- **Alzheimer's disease** is a **mental deterioration** characterized by confusion, memory loss, and other symptoms.
- Alzheimer's disease is caused by the formation of neurofibrillary tangles and **amyloid plaques** in the brain.
- A successful treatment in humans may hinge on early detection of amyloid plaques.
- There is no cure for this disease though some drugs are effective at relieving symptoms.

Microscopic signs of Alzheimer's disease

Amyloid plaque

Neurofibrillary tangle

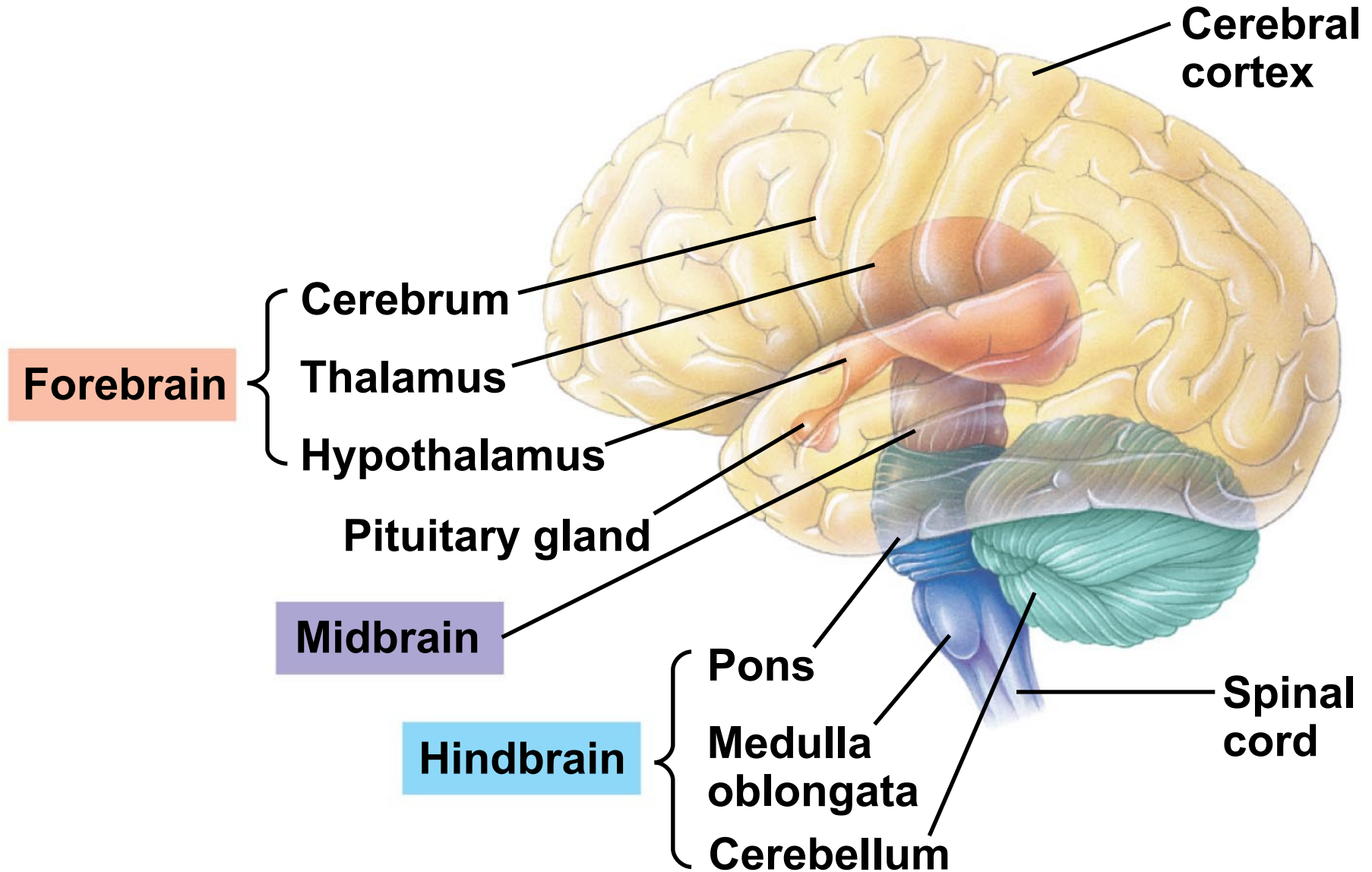
20 μm



Stem Cell–Based Therapy

- Unlike the PNS, the CNS cannot fully repair itself.
- However, it was recently discovered that the adult human brain contains stem cells that can differentiate into mature neurons.
- Induction of stem cell differentiation and **transplantation of cultured stem cells** are potential methods for replacing neurons lost to trauma or disease.

Human Brain



You should now be able to:

1. Compare and contrast the nervous systems of: hydra, sea star, planarian, nematode, clam, squid, and vertebrate.
2. Distinguish between the following pairs of terms: central nervous system, peripheral nervous system; white matter, gray matter; bipolar disorder and major depression.
3. List the types of glia and their functions.
4. Compare the three divisions of the autonomic nervous system.

-
5. Describe the structures and functions of the following brain regions: medulla oblongata, pons, midbrain, cerebellum, thalamus, epithalamus, hypothalamus, and cerebrum.
 6. Describe the specific functions of the brain regions associated with language, speech, emotions, memory, and learning.

-
8. Describe the symptoms and causes of schizophrenia, Alzheimer's disease, and Parkinson's disease
 9. Explain how drug addiction affects the brain reward system